MUKUNDAN MANIVEL

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PROFILE

Chemical Engineering graduate (B.E., 2024) and M.Sc. Advanced Chemical Engineering candidate at the University of Greenwich (2025 – 2026), with a First-Class equivalent record (OGPA 8.68). Combines hands-on process and plant experience with strengths in process simulation (ASPEN Plus and HYSYS) and applied machine learning, and is currently volunteering on an energy research project in Medway.

Objective: To secure a graduate engineering role in process engineering, sustainable energy, chemical manufacturing, or AI-driven engineering, applying process optimisation, simulation, and data skills to deliver safe, efficient, and sustainable operations.

EDUCATION

University of Greenwich — M.Sc. Advanced Chemical Engineering **2025 – 2026**
- Class Representative. Medway, UK.

Annamalai University — B.E. Chemical Engineering **2020 – 2024**
- OGPA 8.68 / 87.18% (First Class equivalent).
- Core modules: Process Design, Mass Transfer Operations, Thermodynamics, Reaction Engineering.

WORK EXPERIENCE

Conserve Engineering, Engineers for Tomorrow **Sep 2024 – Sep 2025**
Process Engineering
- Engaged in comprehensive process engineering training, understanding workflow and documentation standards.
- Developed familiarity with professional Process Design methodologies and engineering documentation standards.

INTERNSHIPS

Panov Organics Pvt Ltd — Chemical Process Operations Intern **Jun 2024 – Aug 2024**
- Gained practical exposure to manufacturing processes and quality control in an organic chemical production plant.
- Analysed and documented process deviations, strengthening familiarity with industrial documentation standards.

TKM Institute of Technology — Onsite Research Scholar **Jun 2023 – Aug 2023**
- Contributed to onsite research including data collection and preliminary analysis under faculty supervision.
- Collaborated with senior faculty on experimental design and research documentation, building teamwork skills.

Sai Life Sciences — Chemical Process Operations Intern **Dec 2022 – Feb 2023**
- Applied chemical process operation principles in an industrial setting, gaining insight into large-scale batch processing.
- Adapted to new standard operating procedures while adhering to strict quality and safety protocols.

IEI (Institution of Engineers, India) — PI Control Intern (Online) **Mar 2022 – Nov 2022**
- Studied Proportional-Integral (PI) control systems for process automation and stability.
- Applied control-loop optimisation principles to maintain efficient and stable process operation.

PROJECTS & RESEARCH

AI-Based Predictive Maintenance for Filter Systems

- Built an ML system on real sensor data (pressure, flow, temperature) to forecast filter degradation and optimal replacement intervals.
- Implemented GAN and diffusion model pipelines in Python (TensorFlow/Keras) to augment a limited operational dataset.
- Enabled condition-based maintenance over fixed schedules, improving filter lifespan and reducing unplanned downtime and cost.

Steam Splitting for Hydrogen/Oxygen Production — Final Year Project

- Designed a Nickel-Alumina catalysed steam-splitting process for hydrogen and oxygen production.
- Simulated process parameters to enhance diesel engine efficiency using the generated products.

Bio-Hydrogen Production from Food Waste

- Researched sustainable bio-hydrogen production from food waste using sedimented mangrove mycoflora (Pichavaram, Tamil Nadu).
- Presented findings in two national conference papers (NIT Warangal; Vel Tech / IIChE Chennai).

PUBLICATIONS

Journal Article — Chemical Engineering Science (Co-author) **Submitted 2026**

- Physics-informed synthetic data generation for remaining useful life (RUL) prediction in industrial filtration systems. Manuscript ID CES-D-26-05836 (under review).

VOLUNTEERING EXPERIENCE

Energy Research Project — Medway, UK **Sep 2026 – Present**

Research Volunteer

- Volunteering on an energy research project, contributing to experimental work, data collection, and analysis.

CERTIFICATIONS & AWARDS

- NPTEL — Physico-chemical Processes for Wastewater Treatment; Organic Chemical Technology.
- 3rd Prize, Ideathon, Annamalai University; ASPEN Plus Simulation workshop, NIT Trichy.
- Two national conference papers on Bio-Hydrogen Production (NIT Warangal; IIChE Chennai).

SKILLS & SOFTWARE

Process Engineering	Process Design · Mass & Heat Transfer · Thermodynamics · Reaction Engineering · Process Optimisation · Quality Control · PI Control
Simulation & CAD	ASPEN Plus · Aspen HYSYS · PIPENET · AutoCAD
Programming & Data	Python · TensorFlow · Keras · Machine Learning · GANs · LSTM/Transformer Models · Git/GitHub
Tools & Professional	Microsoft Office · Problem-Solving · Teamwork · Adaptability · Time Management · Technical Documentation
Design Coordination	Design Reviews · Technical Comment Trackers · Deliverables Trackers · Stakeholder Coordination · Constructability & Operability Review · OPEX